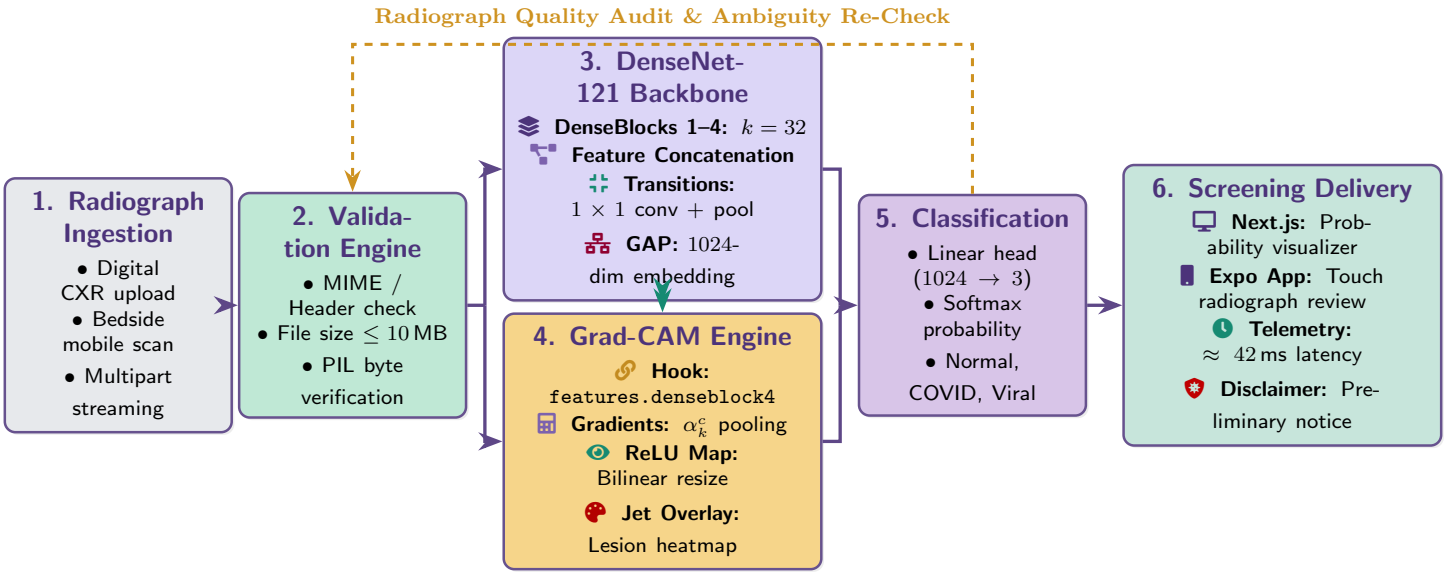

Radiograph Quality Audit & Ambiguity Re-Check
1. Radiograph Ingestion
• Digital CXR upload
• Bedside mobile scan
• Multipart streaming
2. Validation Engine
• MIME / Header check
• File size ≤ 10 MB
• PIL byte verification
3. DenseNet-121 Backbone
DenseBlocks 1–4: k = 32
Feature Concatenation
Transitions: 1 × 1 conv + pool
GAP: 1024-dim embedding
4. Grad-CAM Engine
Hook: features.denseblock4
Gradients: α_k^c pooling
ReLU Map: Bilinear resize
Jet Overlay: Lesion heatmap
5. Classification
• Linear head (1024 → 3)
• Softmax probability
• Normal, COVID, Viral
6. Screening Delivery
Next.js: Probability visualizer
Expo App: Touch radiograph review
Telemetry: ≈ 42 ms latency
Disclaimer: Preliminary notice